Exam Date & Time: 02-May-2025 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

SIXTH SEMESTER B.TECH. (COMPUTER SCIENCE AND ENGINEERING/ INFORMATION TECHNOLOGY) DEGREE EXAMINATIONS - APRIL / MAY 2025
SUBJECT: HUM 3021/HUM_3021/HUM_3151 - ENGINEERING ECONOMICS AND FINANCIAL MANAGEMENT

Answer ALL the questions.
Missing data may be suitably assumed.
Interest factor tables are provided with question and in reference section, for others use formula.
Formula book is provided in the reference section.

ENGINEERING ECONOMICS AND FINANCIAL MANAGEMENT [HUM 3021]

Marks: 50 **Duration: 180 mins.**

A

Answer all the questions.

1A) A loan of ₹20,000 requires monthly payments of ₹526 over a 48-month period of time. These payments include both principal and interest.
i) What is the nominal interest rate for this loan?
ii) What is the effective interest rate per year? (4)
iii) Determine the amount of unpaid loan principal after 24 months.

1B) The following data refers to the cash flows of five investment proposals. The total money available to the company to invest is ₹35,000/-. The company can accept any one of the proposals with the same letter or a combination of proposals with different letters subject to the budget available. Select the best alternative based on the present worth of the total investment criterion. The rate of return is 10%.

Proposal	First Cost	Net Annual Income (years 1-10)
A1	10,000	2,000
A2	12,000	2,100
B1	20,000	3,100
B2	30,000	5,000
C1	35,000	4,500

(3)

10%

n	F/P	P/F	A/F	A/P	F/A	P/A	A/G	P/G
1	1.100	.9091	1.0000	1.1000	1.000	0.909	0	0
2	1.210	.8264	.4762	.5762	2.100	1.736	0.476	0.826
3	1.331	.7513	.3021	.4021	3.310	2.487	0.937	2.329
4	1.464	.6830	.2155	.3155	4.641	3.170	1.381	4.378
5	1.611	.6209	.1638	.2638	6.105	3.791	1.810	6.862
6	1.772	.5645	.1296	.2296	7.716	4.355	2.224	9.684
7	1.949	.5132	.1054	.2054	9.487	4.868	2.622	12.763
8	2.144	.4665	.0874	.1874	11.436	5.335	3.004	16.029
9	2.358	.4241	.0736	.1736	13.579	5.759	3.372	19.421
10	2.594	.3855	.0627	.1627	15.937	6.145	3.725	22.891
11	2.853	.3505	.0540	.1540	18.531	6.495	4.064	26.396
12	3.138	.3186	.0468	.1468	21.384	6.814	4.388	29.901

1C) A working woman is planning for her retired life. She has 20 more years of service. (3)

She earns ₹50,000 per month. She would like to deposit 10% of her salary at the end of the first year and thereafter she wishes to increase the deposit amount every year by ₹1,000 to the previous year for the next 14 years. At an interest rate of 15%, estimate the total amount at the end of the 15th year.

15%

N	F/P	P/F	A/P	P/A	A/F	F/A	P/G	A/G
1	1.150	.8696	1.1500	.870	1.0000	1.000	0	0
2	1.323	.7561	.6151	1.626	.4651	2.150	.756	.465
3	1.521	.6575	.4380	2.283	.2880	3.473	2.071	.907
4	1.749	.5718	.3503	2.855	.2003	4.993	3.786	1.326
5	2.011	.4972	.2983	3.352	.1483	6.742	5.775	1.723
6	2.313	.4323	.2642	3.784	.1142	8.754	7.937	2.097
7	2.660	.3759	.2404	4.160	.0904	11.067	10.192	2.450
8	3.059	.3269	.2229	4.487	.0729	13.727	12.481	2.781
9	3.518	.2843	.2096	4.772	.0596	16.786	14.755	3.092
10	4.046	.2472	.1993	5.019	.0493	20.304	16.979	3.383
11	4.652	.2149	.1911	5.234	.0411	24.349	19.129	3.655
12	5.350	.1869	.1845	5.421	.0345	29.002	21.185	3.908
13	6.153	.1625	.1791	5.583	.0291	34.352	23.135	4.144
14	7.076	.1413	.1747	5.724	.0247	40.505	24.972	4.362
15	8.137	.1229	.1710	5.847	.0210	47.580	26.693	4.565

2A) An industrial-grade asset was acquired for ₹5,00,000 to support critical production operations. It has an expected useful life of 10 years, with a salvage value anticipated to be ₹1,00,000 at the end of this period. Using the Declining Balance Depreciation Method, calculate the undepreciated capital at the end of the 5th year. Also, determine the depreciation charge for the 7th year. (4)

2B) Two assets, X and Y, are being considered for selection based on their ability to effectively perform a required operational function, each with unique financial implications.

Asset Y requires an initial investment of Rs 34,000, with an anticipated salvage value of Rs 4,500 at the end of a 4-year service life with operating expenses of Rs. 1,000 per year.

Asset X, on the other hand, has a lower initial cost, priced Rs 9,500 less than Y, yet has a shorter economic lifespan by one year and no salvage value. Furthermore, X incurs additional annual operating expenses of Rs 2,800 compared to Y. Given a required rate of return of 15%, using study period of 3 years, determine which asset offers the more favourable financial outcome. Assume, the market value is equal to book value and assets depreciate according to straight line method. (3)

15%

n	F/P	P/F	A/F	A/P	F/A	P/A	A/G	P/G
1	1.150	.8696	1.0000	1.1500	1.000	0.870	0	0
2	1.322	.7561	.4651	.6151	2.150	1.626	0.465	0.756
3	1.521	.6575	.2880	.4380	3.472	2.283	0.907	2.071
4	1.749	.5718	.2003	.3503	4.993	2.855	1.326	3.786
5	2.011	.4972	.1483	.2983	6.742	3.352	1.723	5.775

2C) Type 'A' bridge or Type 'B' bridge are under consideration which can be built as a part of a project for the same purpose. The initial cost of a type A bridge is $75,000 and the annual maintenance costs will average $400 per year and last forever. Type B bridge, which must be replaced every 30 years, will cost $40,000 to build and an annual maintenance cost of $700. Both bridges have no salvage value. Using capitalized cost approach with an interest rate is 6%, which alternative is to be selected? (3)

6%

n	F/P	P/F	A/F	A/P	F/A	P/A	A/G	P/G
1	1.060	.9434	1.0000	1.0600	1.000	0.943	0	0
2	1.124	.8900	.4854	.5454	2.060	1.833	0.485	0.890
3	1.191	.8396	.3141	.3741	3.184	2.673	0.961	2.569
4	1.262	.7921	.2286	.2886	4.375	3.465	1.427	4.945
5	1.338	.7473	.1774	.2374	5.637	4.212	1.884	7.934
6	1.419	.7050	.1434	.2034	6.975	4.917	2.330	11.459
7	1.504	.6651	.1191	.1791	8.394	5.582	2.768	15.450
8	1.594	.6274	.1010	.1610	9.897	6.210	3.195	19.841
9	1.689	.5919	.0870	.1470	11.491	6.802	3.613	24.577
10	1.791	.5584	.0759	.1359	13.181	7.360	4.022	29.602
11	1.898	.5268	.0668	.1268	14.972	7.887	4.421	34.870
12	2.012	.4970	.0593	.1193	16.870	8.384	4.811	40.337
13	2.133	.4688	.0530	.1130	18.882	8.853	5.192	45.963
14	2.261	.4423	.0476	.1076	21.015	9.295	5.564	51.713
15	2.397	.4173	.0430	.1030	23.276	9.712	5.926	57.554
16	2.540	.3936	.0390	.0990	25.672	10.106	6.279	63.459
17	2.693	.3714	.0354	.0954	28.213	10.477	6.624	69.401
18	2.854	.3503	.0324	.0924	30.906	10.828	6.960	75.357
19	3.026	.3305	.0296	.0896	33.760	11.158	7.287	81.306
20	3.207	.3118	.0272	.0872	36.786	11.470	7.605	87.230
21	3.400	.2942	.0250	.0850	39.993	11.764	7.915	93.113
22	3.604	.2775	.0230	.0830	43.392	12.042	8.217	98.941
23	3.820	.2618	.0213	.0813	46.996	12.303	8.510	104.700
24	4.049	.2470	.0197	.0797	50.815	12.550	8.795	110.381
25	4.292	.2330	.0182	.0782	54.864	12.783	9.072	115.973
26	4.549	.2198	.0169	.0769	59.156	13.003	9.341	121.468
27	4.822	.2074	.0157	.0757	63.706	13.211	9.603	126.860
28	5.112	.1956	.0146	.0746	68.528	13.406	9.857	132.142
29	5.418	.1846	.0136	.0736	73.640	13.591	10.103	137.309
30	5.743	.1741	.0126	.0726	79.058	13.765	10.342	142.359

3A) A company (which sets MARR = 10%) must purchase new production equipment and is considering two alternatives.

Option 1: Computer controlled	**Option 2: Manual controlled**
First cost: $2,000,000	First cost: $1,300,000
Predicted life: 10 years	Predicted life: 07 years
Salvage value: $150,000	Salvage value: none
Labor cost: $150 per hour	Labor cost: $195 per hour

Determine the annual hours of operation, which makes these two alternatives equivalent. (3)

10%

n	F/P	P/F	A/F	A/P	F/A	P/A	A/G
1	1.100	.9091	1.0000	1.1000	1.000	0.909	0
2	1.210	.8264	.4762	.5762	2.100	1.736	0.476
3	1.331	.7513	.3021	.4021	3.310	2.487	0.937
4	1.464	.6830	.2155	.3155	4.641	3.170	1.381
5	1.611	.6209	.1638	.2638	6.105	3.791	1.810
6	1.772	.5645	.1296	.2296	7.716	4.355	2.224
7	1.949	.5132	.1054	.2054	9.487	4.868	2.622
8	2.144	.4665	.0874	.1874	11.436	5.335	3.004
9	2.358	.4241	.0736	.1736	13.579	5.759	3.372
10	2.594	.3855	.0627	.1627	15.937	6.145	3.725

3B) Chem-Tex Chemical is considering two additives for improving the dry-weather stability of its low-cost acrylic paint. (3)
Additive A has a first cost of $110,000 and an annual operating cost of $60,000.
Additive B has a first cost of $175,000 and an annual operating cost of $35,000.
If the company uses a 3-year recovery period for paint products and a MARR of 20% per year, which process is economically favoured? Use Incremental ROR analysis.

20%

n	F/P	P/F	A/F	A/P	F/A	P/A	A/G	P/G
1	1.200	.8333	1.0000	1.2000	1.000	0.833	0	0
2	1.440	.6944	.4545	.6545	2.200	1.528	0.455	0.694
3	1.728	.5787	.2747	.4747	3.640	2.106	0.879	1.852
4	2.074	.4823	.1863	.3863	5.368	2.589	1.274	3.299
5	2.488	.4019	.1344	.3344	7.442	2.991	1.641	4.906
6	2.986	.3349	.1007	.3007	9.930	3.326	1.979	6.581
7	3.583	.2791	.0774	.2774	12.916	3.605	2.290	8.255
8	4.300	.2326	.0606	.2606	16.499	3.837	2.576	9.883

3C) Two possible routes for laying a power line are under study. Data on the routes are as follows:

	Around the lake	Under the lake
Length	15 km	5 km
First Cost (₹)	1,50,000 per km	7,50,000 per km
Useful life (Years)	15	15
Maintenance Costs (₹)	6,000 /km/year	12,000 /km/year
Salvage Value (₹)	90,000 per km	1,50,000 per km
Yearly power loss (₹)	15,000 per km	15,000 per km

If 15% interest is used, should the power line be routed around the lake or under the lake? Use Annual worth method. (4)

15%

n	F/P	P/F	A/F	A/P	F/A	P/A	A/G	P/G
1	1.150	.8696	1.0000	1.1500	1.000	0.870	0	0
2	1.322	.7561	.4651	.6151	2.150	1.626	0.465	0.756
3	1.521	.6575	.2880	.4380	3.472	2.283	0.907	2.071
4	1.749	.5718	.2003	.3503	4.993	2.855	1.326	3.786
5	2.011	.4972	.1483	.2983	6.742	3.352	1.723	5.775
6	2.313	.4323	.1142	.2642	8.754	3.784	2.097	7.937
7	2.660	.3759	.0904	.2404	11.067	4.160	2.450	10.192
8	3.059	.3269	.0729	.2229	13.727	4.487	2.781	12.481
9	3.518	.2843	.0596	.2096	16.786	4.772	3.092	14.755
10	4.046	.2472	.0493	.1993	20.304	5.019	3.383	16.979
11	4.652	.2149	.0411	.1911	24.349	5.234	3.655	19.129
12	5.350	.1869	.0345	.1845	29.002	5.421	3.908	21.185
13	6.153	.1625	.0291	.1791	34.352	5.583	4.144	23.135
14	7.076	.1413	.0247	.1747	40.505	5.724	4.362	24.972
15	8.137	.1229	.0210	.1710	47.580	5.847	4.565	26.693

4A) A car purchased six years ago for ₹5.50 lakhs by an ABC Ltd. can be sold now for ₹3.5 lakhs or can be kept for another five years. In that case, it will have an annual operation and maintenance cost of ₹50,000 for the first year of operation from now and increases every year by an amount of 20,000. It is expected to have a salvage value of ₹2.25 Lakhs, ₹2 Lakhs, ₹1.75 Lakhs, ₹1.5 Lakhs, ₹1.25 Lakhs for years seven to eleven respectively. A suitable replacement a new car is priced at ₹8.25 Lakhs and an annual operation and maintenance cost of ₹30,000. The salvage value of the new car is ₹5 Lakhs after six years of its life. Evaluate and decide whether the old car can be replaced (Risk adjusted MARR=10%). (4)

10%

n	F/P	P/F	A/F	A/P	F/A	P/A	A/G	P/G
1	1.100	.9091	1.0000	1.1000	1.000	0.909	0	0
2	1.210	.8264	.4762	.5762	2.100	1.736	0.476	0.826
3	1.331	.7513	.3021	.4021	3.310	2.487	0.937	2.329
4	1.464	.6830	.2155	.3155	4.641	3.170	1.381	4.378
5	1.611	.6209	.1638	.2638	6.105	3.791	1.810	6.862
6	1.772	.5645	.1296	.2296	7.716	4.355	2.224	9.684
7	1.949	.5132	.1054	.2054	9.487	4.868	2.622	12.763

4B) (3)

The Trading and Profit and Loss Account of ABC Ltd. for the year 31-3-2021 is given below:

Particulars	Rs	Particulars	Rs
To opening stock	76,250	By Sales	5,00,000
To purchases	3,15,250	By Closing stock	98,500
To Carriage inward	2,000		
To wages	5,000		
To Gross profit c/f	2,00,000		
	5,98,500		**5,98,500**
To Administrative exp.	1,01,000	By Gross profit b/d	2,00,000
To Selling & dist. Exp.	12,000	By interest on securities	1,500
To non-operating exp.	2,000	By dividend on shares	3,750
To financial exp.	7,000	By profit on sale of shares	750
To net profit c/d	84,000		
	2,06,000		**2,06,000**

From the above information determine:

a) Gross Profit Ratio

b) Net Profit Ratio

c) Stock Turnover Ratio

4C) The manager of Erie Pvt. Ltd. is involved in a production planning process and wishes to conduct a risk assessment using two scenarios, the worst case and best-case scenarios, based on the company's current estimated levels. Based on the following information about Erie Ltd, you must conduct a break-even analysis under these scenarios. The manager adopts the pricing strategy that involves determining the selling price of the product and then figuring out if it can be made at a cost lower than that.

i) Under best case scenario the manager expects the selling price of ₹47,000, and the variable cost of manufacturing the unit is ₹27,200. What could the company's fixed cost per year be to break even with sales of 18,000 units per year? (3)

ii) Under the worst-case scenario, the manager expects the selling price dropping to ₹36,000 and variable cost increasing to ₹30,000, in this situation what would be the fixed cost per year to break even if the sales remain unchanged at 18,000 units per year.
Based on the results of the break-even analysis for both the Best-Case and Worst-Case scenarios, how do these help in the company's financial risk minimization and decision-making?

5A) Why is the current ratio considered a key liquidity ratio in assessing a company's financial health? How does it differ from the quick ratio, and why is the distinction important? (2)

5B) A frozen fish company is planning an expansion to a cold storage facility. Two alternative site design proposals are being considered at an interest rate of 10% after consideration of risk in investments. Plant A requires an expenditure of $350,000 for land, while Plant B requires $425,000. Assume that the land will be sold after 30 years at the same purchase price. The buildings are expected to last 30 years, the compressors and related equipment will last 10 years before requiring replacement. The annual maintenance of the any (4)

compressor costs $1,000 and increases by $500 every year throughout the compressor's life.
The energy costs of the facility are as given below and will increase throughout the building's life. Neither the buildings nor the equipment is expected to have any salvage value. With this information and the data provided below make an annual cost comparison to determine which proposal is preferred.

	A	B
Building and insulation, $	600,000	400,000
Compressors, $	100,000	85,000
Expected energy costs in First year, $	65,000	65,000
Energy Cost Increase for each additional year, $	3,000	4,000
Annual maintenance expense, $/year	20,000	50,000

10%

n	F/P	P/F	A/F	A/P	F/A	P/A	A/G	P/G
1	1.100	.9091	1.0000	1.1000	1.000	0.909	0	0
2	1.210	.8264	.4762	.5762	2.100	1.736	0.476	0.826
3	1.331	.7513	.3021	.4021	3.310	2.487	0.937	2.329
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7	1.949	.5132	.1054	.2054	9.487	4.868	2.622	12.763
8	2.144	.4665	.0874	.1874	11.436	5.335	3.004	16.029
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12	3.138	.3186	.0468	.1468	21.384	6.814	4.388	29.901
13	3.452	.2897	.0408	.1408	24.523	7.103	4.699	33.377
14	3.797	.2633	.0357	.1357	27.975	7.367	4.996	36.801
15	4.177	.2394	.0315	.1315	31.772	7.606	5.279	40.152
28	14.421	.0693	.00745	.1075	134.210	9.307	7.914	73.650
29	15.863	.0630	.00673	.1067	148.631	9.370	8.049	75.415
30	17.449	.0573	.00608	.1061	164.494	9.427	8.176	77.077

5C) An equipment having a life of 15 years with an initial cost of $12,100 is being considered for replacement The salvage value of a one-year-old equipment is $7,000 and this value then onwards decreases every year by an amount of $300.The maintenance and operating cost is $1,200 in the first year and then onwards it is expected to increase by $1,000 every year in future years. Compute the economic life of this asset and the minimum EUAC, with i=10%.

10%

n	F/P	P/F	A/F	A/P	F/A	P/A	A/G	P/G
1	1.100	.9091	1.0000	1.1000	1.000	0.909	0	0
2	1.210	.8264	.4762	.5762	2.100	1.736	0.476	0.826
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12	3.138	.3186	.0468	.1468	21.384	6.814	4.388	29.901
13	3.452	.2897	.0408	.1408	24.523	7.103	4.699	33.377
14	3.797	.2633	.0357	.1357	27.975	7.367	4.996	36.801
15	4.177	.2394	.0315	.1315	31.772	7.606	5.279	40.152

(4)

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